

PaperCut Software: Delivering a cloud-based printing analytics product

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Google Cloud results

- Enabled the business to reduce the cost of PaperCut Views by 30 percent
- Cut the time required to push updates from 30 minutes to a few minutes
- Gave business a platform that allowed them to avoid the cost of full time Dev personnel

PaperCut Software provides print management services ranging from tracking and reporting on print jobs to integrated printing and device management in corporate environments. The business custom job management. The b



About PaperCut

Founded in 1998, PaperCut Software is a privately owned and operated software product company. With offices in Melbourne, Australia, Portland, Oregon, and London, UK, the fast-growing and quirky company is dedicated to supporting customers as they look to save paper and reduce the costs associated with printing. PaperCut has nearly two decades of experience in print management software and has deployed their software in more than 50,000 organizations worldwide, tracking the printing and copying of over 65 million users.

protect the environment by reducing paper used for printing.

Industries: Technology

Location: Australia

Founded in 1998, PaperCut Software significantly reduced paper usage each year over the last decade. PaperCut Software now has over 50,000 customers in over 100 countries. These customers include organizations such as Oxford University, Harvard University, the University of California, and Starbucks. "We are proud to be part of this success story, especially considering that the majority of our sales come from North America and Europe," says Chris Dance, CEO and Officer and co-founder, PaperCut Software.

Technology is integral to PaperCut Software. Chris Dance believes the business is best placed to move from a traditionally staid area of IT to a more dynamic one. "We stand among the world's most innovative and disruptive software companies," he adds. "We have a strong community of people around the world who are passionate about our software. We aim to help make their lives easier over the long term."

Dataflow (<https://cloud.google.com/dataflow/>),

Cloud Datastore

(<https://cloud.google.com/datastore/>)

BigQuery (<https://cloud.google.com/bigquery/>),

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>),

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

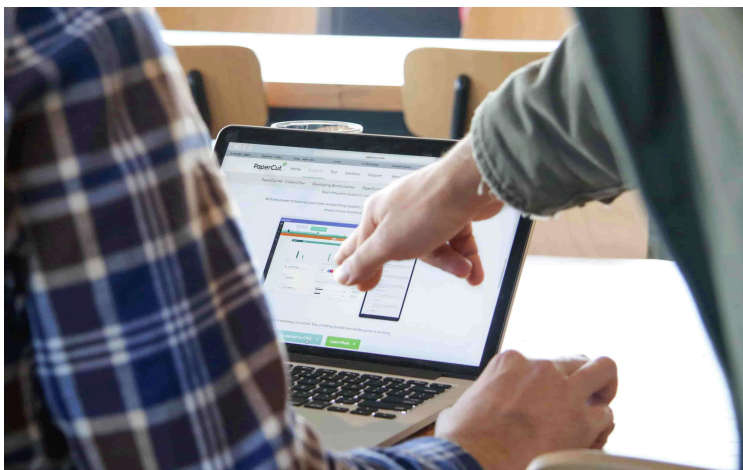
App Engine (<https://cloud.google.com/appengine/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Operations

(<https://cloud.google.com/products/operations>)



Challenge

PaperCut Software saw an opportunity to augment its flagship PaperCut MF product—which provides print, copy, scan, and fax control through software embedded in clients' on-premises devices—with a cloud-based print logging, benchmarking, and comparison service. "More broadly, we want to adopt the cloud to stay on the edge as a technology company and remain relevant to our customers," says Dance.

To achieve this, PaperCut began developing PaperCut Views, a free cloud-based service providing printing analytics and insights for small businesses and home offices, and started running it in a public cloud infrastructure. However, a discussion about where the business wanted to be and what the software development landscape may look like in five years prompted Dance and co-founder Matt Doran to consider options for the future.

Solution

"We believed that Google's Platform as a Service approach would ultimately win out over Infrastructure as a Service," says Dance. "In addition, when we delved into the specifics of services such as the Google BigQuery data analytics warehouse, we found they were market leaders in terms of functionality and performance."

PaperCut began its Google Cloud deployment with some internal pilot projects. These projects involved deploying internal licensing and customer management databases on App Engine and

Compute Engine. After a year of internal systems work, PaperCut decided to commit PaperCut Views to the platform. Following the decision, the business took nine months to bring the print analysis product to market. An agreement with HP, Inc. to ship PaperCut Views with the vendor's printers provided a fillip to the business and vindicated its new strategic direction.

PaperCut Software is now running PaperCut Views on Google Cloud. This enables businesses to understand their printing, forecast paper and toner usage, and identify areas where they can save paper. The HP, Inc. version of the product has been enhanced to enable toner ordering from online office supplies organizations and has to date been shipped with thousands of the vendor's printers. The agreement with HP, Inc. has enabled PaperCut Software to scale PaperCut Views before providing similar functionality as an option to users of its on-premises products. In April 2017, more than 19,000 customers were running PaperCut Views.

PaperCut Software is taking the lessons from PaperCut Views and investigating how they can make them work with their existing product suite. "Over the next 12 months we hope to get a significant percentage of our on-premise customers choosing to enable our cloud services," Dance says.

PaperCut Views is now tracking about 45,000 printing devices in real time and providing users with critical information such as user printing and equipment error events. Client edge computing queries printers and sends the information to an API running in the Google Cloud. Cloud Pub/Sub or Task

Queues create asynchronous message flows that App Engine listens to and updates Cloud Datastore accordingly. The messages are also streamed into BigQuery for analysis through Cloud Dataflow. Memcache expedites and lowers the cost of query responses.

Benefits

The scalability provided by Google Cloud has enabled PaperCut Software to support demand peaks of up to 10 times average traffic for PaperCut Views without compromising sub-second response times or near-perfect availability. Google Cloud has also provided the security needed to win confidence from customers in the new cloud product. "Security and vulnerability scanning is an automated feature and avoids us having to hire or procure the services of security experts in our own company," says Dance.



Reduced costs, lowered time to push out builds

Working with Google Cloud has not just delivered technical and business benefits to PaperCut Software. "The biggest advantage for us as a business is that Google Cloud is a modern platform and a modern way of thinking," says Dance. "This is flowing into our culture—not just the development teams working in the cloud, but the teams working on on-premises software."

By undertaking detailed analysis and using monitoring tools such as Google Stackdriver, PaperCut Software has been able to make changes to its software to reduce the operating costs for PaperCut Views by about 30 percent. In addition, by moving its build environment onto Container Engine, the business has been able to slash the time required to push each build out from 30 minutes on average to a few minutes. Furthermore, PaperCut Software has also been able to limit its investment in DevOps experts to one person on the system administration side, rather than also being required to recruit similarly qualified experts on the development and production sides.

PaperCut Software is now taking into Google Cloud a service that enables customers to manage printing from bring-your-own devices, as well as its internal systems that manage client licensing and ordering.

With Google Cloud providing the platform, PaperCut Software is poised to move components of its on-premises products to the cloud sooner rather than later. "Google Cloud has given us the confidence to build our business and work with high calibre partners and we look forward to continue working

with Google to expand into new markets in future,"
says Dance.